

ARQAM AHMED ANSARI

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EXPERIENCE

Genetech Solutions | AI Engineer

June 2025 – Present | On-Site

- Built an agentic chatbot for the Genetech website using LangGraph for agent orchestration, FastAPI for the backend, and React for the frontend, with Dockerized deployment on an on-premises server and LangSmith for observability. Key features:
 - Short-term memory for context-aware conversations
 - User journey tracking across the website with personalized popup messages
 - Forms connecting visitors directly to sales and consultation teams
 - All user conversations logged to the CRM for bot retraining
 - Leads captured by the bot automatically synced to the CRM
- Built the Islamic Knowledge Hub, a local agentic system that answers users' Islamic questions based on Islamic books, using specialized sub-agents for each category:
 - LangGraph orchestration with short-term memory
 - Local inference on Qwen 3.5 14B (4-bit) via vLLM, running on an RTX 3090 Ti GPU for high throughput
 - BAAI/bge-M3 embedding model with rerankers, plus RAG evaluation
 - Qdrant vector database with metadata-separated books for filtered retrieval
- MedBook AI: Voice-Based Hospital Doctor Booking Agent
 - Voice Pipeline: Built a LiveKit voice agent powered by Gemini's gemini-3.1-flash-live-preview live voice-to-voice model, with GPT orchestrating tool-calling and conversation logic.
 - Tool-Calling Orchestration: LLM orchestrates 10+ LiveKit tools that connect to each doctor's calendar to check availability and book appointments live on the call.
 - Memory: Integrated Mem0 for conversational memory across calls.
 - Confirmations: Automatically sends confirmation emails to both doctor and patient after a booking is made.
 - Mobile Deployment: Deployed the agent as a cross-platform Flutter app (Android & iOS) using the LiveKit Flutter SDK for on-phone voice booking.
- Automating Product Engineering (SDLC → ADLC)
 - Automated the software development lifecycle into an AI-driven development lifecycle (ADLC) spanning design to production, saving ~60% time across PM, Design, Frontend, Backend, SQA, and DevOps teams.
 - Integrated Codex, Claude, and Kombar into the design team's workflow; connected a Figma MCP server with custom skills to generate pixel-perfect frontend code directly from Figma frames.
 - Integrated a Context7 MCP server for backend code generation and best practices, a Test Script MCP for automated unit testing across frontend and backend, and custom skills with a SonarQube MCP server to automate security and reliability audits within coding harnesses.
 - Integrated Playwright MCP and Chrome DevTools MCP to automate SQA, detect bugs, and automatically report them to the CRM with developer assignment.
 - Connected all MCP servers in a sandboxed environment within Claude Project folders, combining plugins, skills, and README-based deployment/project guidance to help DevOps run audit logs across multiple simultaneous projects.
 - Orchestrated the entire system into a unified "Dark Factory" — an end-to-end ADLC pipeline that autonomously builds, validates, tests, and deploys.

Arcana Info | AI and Automation MTO

April 8, 2025 | On-Site

- Developed a Captcha-solving model integrated into RPA for Bank Alfalah, enabling automated task execution.
- Used Denodo for data virtualization in a pharma project, powering BI dashboard creation.

PROJECTS

Fin GPT: Financial Research RAG on Annual Reports

- **Challenge:** Built a RAG system over a client's 20+ years of annual reports, where key information was embedded in charts and complex tables.
- **Hybrid Ingestion Pipeline:** Used the Marker and Opendataloader libraries to separate text and images and convert PDFs to Markdown (text and tables), while diagrams were sent to a vision-language model for description generation.
- **Structure-Preserving Chunking:** Applied hierarchical chunking with a separate table-aware chunker (Chunkie library) to preserve table structure.

- **Retrieval:** Ingested into a vector database supporting hybrid lexical (BM25) and semantic search, combined with reciprocal rank fusion and rerankers.
- **Memory:** Implemented short-term and long-term memory (Mem0), with metadata-based filtering across 20+ years of filings and PostgreSQL as the long-term memory checkpointer store.

Harness Eng: Autonomous Equity Research Agent

Multi-Agent LLM Harness | LangGraph / Deep Agents

- **Multi-Agent Architecture:** Designed and implemented a multi-agent LLM harness on LangGraph/Deep Agents that decomposes a research task across 4 specialized sub-agents (filings analysis, competitive research, transcript analysis, quantitative modeling), each with isolated context and scoped tool access.
- **MCP & Sandboxed Execution:** Integrated external data sources via the Model Context Protocol (MCP), and implemented sandboxed code execution so the modeling sub-agent runs real pandas/numpy valuation code rather than hallucinating numbers.
- **Crash-Resumable Execution:** Implemented persistent, crash-resumable execution using LangGraph's checkpointing layer (SQLite/Postgres-backed), so multi-hour research runs survive process interruption and resume from the exact point of failure.
- **Memory & Skills:** Built a two-tier memory system: long-term firm conventions injected at startup via a persistent memory file, and short-term thread-scoped state for in-session context, plus progressive-disclosure "skills" so domain methodology (DCF conventions, red-flag checklists) is loaded only when relevant, keeping the main context window lean.

TECHNICAL SKILLS

Programming: Python, C++, Java

Database: SQL, MongoDB, Pinecone, Redis, Qdrant, FAISS, Chroma

ML/DL: Scikit-learn, TensorFlow, PyTorch

MLOps: MLflow (Experiment Tracking), Evidently AI (Data Drift), DVC, GitHub Actions (CI/CD), Docker, Grafana, Prometheus, Kubernetes

RAG: Corrective RAG, Self RAG, Self-Correcting RAG, Graph RAG, Vector-less RAG, File-based RAG, Hybrid/Ensemble RAG

Retrieval Toolkit: Vector Search, BM25, Rerankers, Reciprocal Rank Fusion, Ensemble Retrieval, Docling, MinerU, Opendataloader

LLM Observability: LangSmith, Opik, Custom Observability

Voice Agents: LiveKit, Pipecat, STT, TTS, Voice-to-Voice Models, Deepgram, ElevenLabs, AssemblyAI, Whisper

Mobile: Flutter, Dart, LiveKit Flutter SDK

AI Harness Engineering: Deep Agents, Sub-Agent Architectures, Sandboxes, Checkpointers, Long-Term Memory, Short-Term Memory

Agentic AI Frameworks: LangChain, LangGraph, CrewAI, Mem0

LLM Inference: Ollama, vLLM, llama.cpp

Deployment: Flask, AWS Elastic Beanstalk, AWS SageMaker, AWS EC2, AWS ECR, Nginx, PM2, Linux

EDUCATION

University of Karachi

BS in Computer Science

2021 – 2024 | Karachi, Pakistan

CERTIFICATIONS

- Udemy – Python OOP
- NVIDIA – Accelerated Computing in CUDA C/C++
- ACE Online Courses – Comprehensive AI/ML Course